Kick-Starting Virtualisation with VirtualBox

This article looks at some quick steps for installing, configuring and working with Oracle VM VirtualBox.

Virtualisation has been growing in popularity ever since it first appeared on the scene in the early 1960s, as an effective and reliable way to share compute resources across mainframes by the creation of multiple virtual machines, each running with their own defined RAM and disk space. Virtualisation software adoption gained impetus from the year 2005 onwards and has grown ever since, faster than anyone (including the experts) imagined. There are three main areas of IT where virtualisation is currently making inroads—network virtualisation, storage virtualisation and server virtualisation; but for this article we are going to look at a much simpler version called Operating System Virtualisation.

Operating System Virtualisation enables users to create multiple guest operating systems (OS) within a single 'host' OS (see Figure 1). The host OS is the one that you see when you start your computer. Each guest OS is provided with some definite RAM and CPU from the host machine, for its consumption. Such virtualisation is useful when you are aiming to do research, testing or development work, where virtual machines can be flexibly provisioned, used, and then deleted after some time. This actually helps save time and the cost of procuring unnecessary hardware.

There are a lot of open source as well as commercial OS virtualisation software on the market, each with its own pros and cons. Here's a quick comparison between some of the most widely used OS virtualisation products: QEMU, Oracle VM VirtualBox, VMware Player and Parallels Desktop 8 for Mac.

Introduction to VirtualBox

VirtualBox is a cross-platform virtualisation software designed to run on most modern x86 systems. Initially developed by a company named Innotek, VirtualBox was bought over by Sun Microsystems in 2008 and is now developed by Oracle Corporation as part of its family of virtualisation products. VirtualBox is installed on an existing host OS as a simple application. You can then create additional virtual machines (VMs) with this, which contain guest operating systems. Each guest OS gets its own virtual environment that comprises some dedicated CPU, RAM as well as hard disk space. Besides these, VirtualBox also provides some additional useful features such as: